

Gabriel Tomassini

PHD CANDIDATE IN ASTROPHYSICS

Contact

Observatoire de la Côte d'Azur
Nice, France

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Nationality: French

Skills

Observations
SPHERE/VLT imaging, MATISSE/VLTI interferometry: data reduction & analysis

Modelling
Radiative transfer, SED fitting, 3D radiation-hydrodynamics (CO5BOLD)

Programming
Python, Fortran, C++, HPC code optimisation, LaTeX

Languages

French
Native

English
Advanced (C1)

Spanish
Intermediate (B2)

Interests

Astronomy, computer science

Sports (triathlon, motorsport)

Astrophysicist working on binaries of evolved stars, combining observations, radiative transfer and 3D radiation-hydrodynamics simulations to understand stellar mass loss and the role of companions in the late stages of stellar evolution.

EDUCATION

PhD in Astrophysics Oct 2024 – present
Observatoire de la Côte d'Azur, Université Côte d'Azur, Nice, France

- Developed a public SED-fitting code based on 1D radiative transfer for dusty evolved stars
- Extended the 3D RHD code CO5BOLD to include a companion; restored its vectorisation (×5 speed-up)
- SPHERE/VLT images of AFGL 4106 featured as ESO Picture of the Week (potw2608)

MSc in Astronomy & Astrophysics (Master 2) Sep 2023 – Jul 2024
Observatoire de Paris – PSL University, Paris, France — with honours

Magistère de Physique Fondamentale (BSc + MSc level) Sep 2021 – Jul 2024
Université Paris-Saclay, Orsay, France — with honours
Selective three-year honours programme in fundamental physics

RESEARCH EXPERIENCE

Research Internship (MSc thesis) Mar 2024 – Jul 2024
Observatoire de la Côte d'Azur, Nice — Supervisor: Eric Lagadec

- Analysis of SPHERE/VLT imaging and MATISSE/VLTI interferometric data of the post-AGB binary candidate AFGL 4106
- Comparison with state-of-the-art models of binary interaction

Research Internship May 2023 – Aug 2023
Observatoire de la Côte d'Azur, Nice — Supervisor: Eric Lagadec

- Analysis of SPHERE/VLT observations of AFGL 4106 and comparison with binary-interaction models

Research Internship May 2022 – Jul 2022
Institut d'Astrophysique de Paris, Paris — Supervisor: Robert Mochkovitch

- Simulations of high-energy phenomena associated with binary neutron star mergers

TEACHING

Teaching Assistant in Mathematics and Physics Sep 2025 – present
Université Côte d'Azur, Nice, France

- Tutorial classes for first-year undergraduate students (58 hours per year)

OUTREACH

Science Communicator (part-time, alongside PhD) Nov 2024 – Jul 2025
Observatoire de la Côte d'Azur, Nice, France

- EduCosmos project: designed hands-on astronomy workshops
- Public science events (Fête de la Science, Festival des Sciences Océans, ...)